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Where predictive information resides in endogenous A-to-I RNA editing

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Why one endogenous double-stranded RNA substrate, or one adenosine within it, is edited remains unresolved. We decomposed this selection problem into intrinsic whole-duplex suitability, transcript and cellular realization, target-local selectivity, and the separate statistical dependence among sites from one substrate. Across 884 human inverted-Alu substrates, controlled prediction tested candidate information against a common sequence–structure reference using pair-disjoint development, matched controls, and explicit evidence classes. Whole-duplex sequence and predicted structure supplied broad signal, and genomic organization added a distinct between-substrate propensity. The same sites differed across tissues, yet available RBP, tissue-average abundance, transcript, repeat, and bulk chromatin measurements supplied no stable residual site-level information. Expanded valid labels and pair-first weighting corrected underused supervision and unequal substrate influence. A representation centered on the target and organized by predicted pairing added information beyond whole-duplex pooling; shifting the same-capacity window erased that gain. A parallel search of richer supervision, cross-context sharing, pretrained RNA representations, and greater capacity or integration produced mostly limited, neutral, or adverse returns. Thus biologically and statistically aligned changes recovered signal that generic model sophistication alone did not under the tested regime. A predictor frozen from the supported principles improved AUROC, AUPRC, F1, and Brier score in all five held-out endpoints. The resulting map locates recoverable information at substrate and target scales and identifies the finer matched measurements needed to interrogate the remaining variation.

Significance

Endogenous A-to-I RNA editing requires selection of an RNA duplex and of particular adenosines within it. Which information governs these choices—and which measurements can recover it—remains unclear. We organized a controlled predictive search around intrinsic whole-duplex properties, transcript and cellular context, and target-local selectivity, while treating dependence among sites from one substrate as a separate statistical problem. Sequence–structure, genomic organization, and correctly centered local context supplied reproducible signals; current bulk RBP, transcript, repeat, and chromatin measurements did not resolve residual site-level variation. Biologically and statistically aligned changes improved prediction more than generic model expansion under the tested regime. The remaining gap points toward donor-, cell-, transcript-, regulator-, and structure-matched measurements.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA). The reaction diversifies transcripts and prevents endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Editing is selective at two levels: many potential duplexes show little editing, and neighboring adenosines in one predicted helix can have sharply different outcomes. The same genomic site can also differ among tissues (11, 12).

This selection problem has three biologically distinct scales. Whole-duplex sequence, folding, and genomic organization constrain whether a transcript can form a suitable substrate. Transcript processing, localization, regulator exposure, and enzyme state may determine whether that substrate is realized in a particular cell. Within an available duplex, nucleotide identity, mismatches, bulges, and local helix context influence which adenosine is edited (13–17). These questions require different measurements and controls. They are also distinct from a statistical issue in the benchmark: sites from one duplex are dependent observations, so site-rich substrates can exert disproportionate influence during learning.

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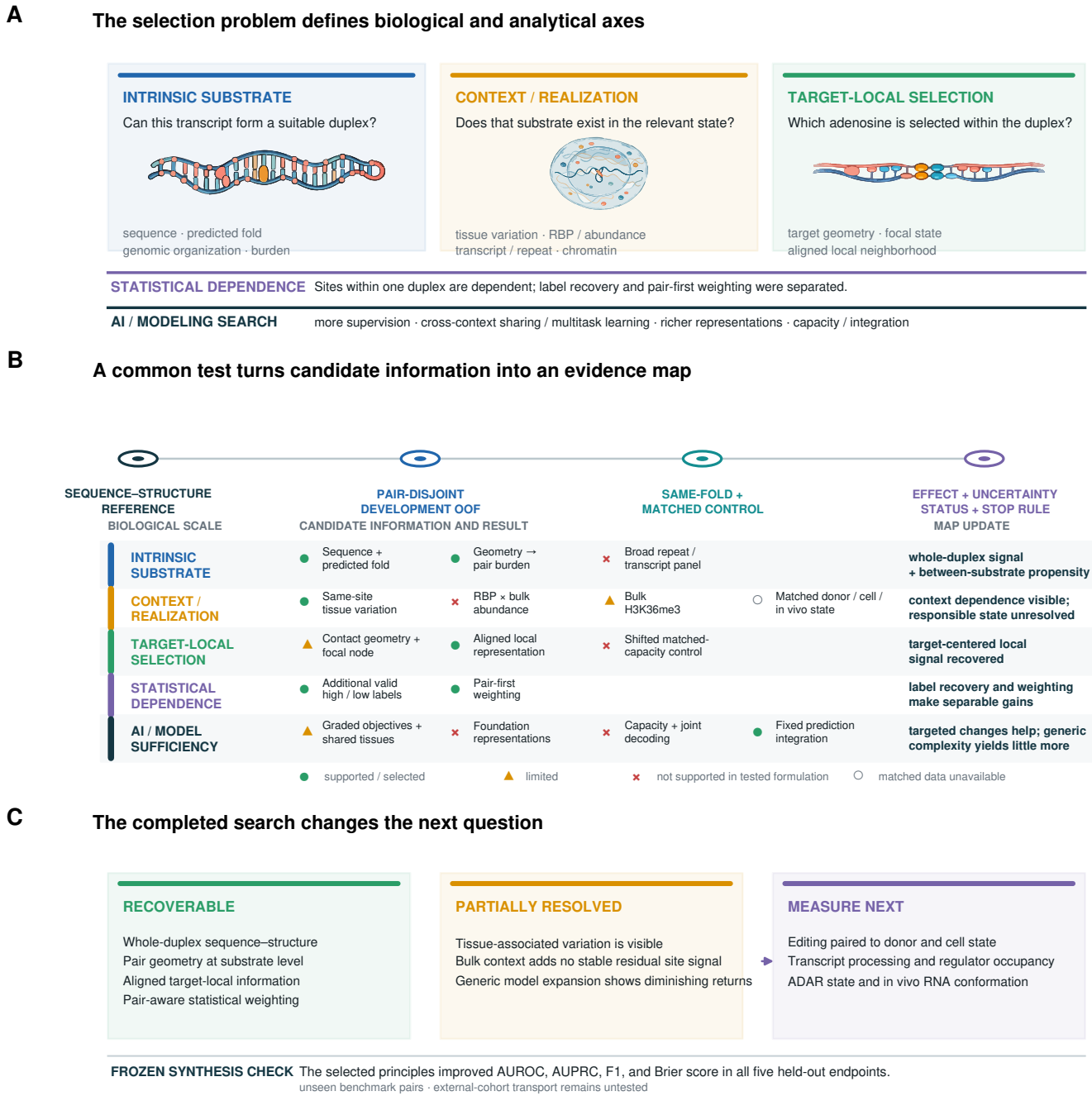


Fig. 1. Systematic search across the information space accessible to the benchmark. (A) The biological selection problem separates into intrinsic whole-duplex suitability, transcript and cellular realization, and target-local adenosine selection; dependence among sites and the ability of richer modeling to close the gap form separate analytical axes. Illustrations are conceptual. (B) Candidate measurements and representations entered a common sequence–structure reference, pair-disjoint development comparison, and matched-control logic. Symbols distinguish supported or selected, limited, unsupported in the tested formulation, and deferred branches. (C) The resulting map separates information recoverable at current resolution from context that remains only partially resolved and measurements that require matched donor, cell, transcript, regulator, and in vivo structure data. The frozen held-out evaluation tested only the selected synthesis on unseen benchmark pairs. SI Appendix, Table S10 gives the complete branch-level coverage and disposition.

The scale decomposition defines a tractable search over the information accessible to the benchmark. Controlled held-out prediction cannot establish mechanism, but it can ask whether a measurement or representation adds reproducible signal beyond a fixed reference. A positive increment locates recoverable information absent from that reference. A null result constrains only the tested measurement and formulation; redundancy, noise, finite data, or model misspecification remain possible explanations. The combination of matched comparisons, falsification controls, transfer tests, and conditional saturation therefore maps empirical recoverability without equating prediction with causality.

We organized biological candidates by scale rather than by model family (Fig. 1). A parallel model-sufficiency axis tested whether richer supervision, cross-context sharing, pretrained representations, or greater capacity and integration could close the residual gap. Branches were compared out of fold against the appropriate common reference and controls; the held-out test was reserved for the frozen synthesis. The Methods and SI Appendix, Table S10 specify the search and evidence rules.

AdarEdit established that sequence and predicted structure can predict site-level editing in this pair-partitioned benchmark (18). Here that model provides the common reference for a different question: which information scales account for the recoverable signal, what adds to it, where plausible measurements fail, and when added modeling sophistication yields diminishing returns. The search finds information at whole-substrate and target-local scales, detects tissue-associated variation that present bulk measurements do not resolve, and identifies pair-first weighting as a separate correction for dependent observations. The selected predictor is the final test of whether those supported principles combine on unseen benchmark pairs.

Results

Search design separates discovery from held-out evaluation. Figure 1 defines the biological, statistical, and model-sufficiency axes and records which branches were supported, limited, not supported, superseded, or deferred. Donor-resolved cellular state and in vivo RNA conformation remain outside the benchmark's measurement scope; their absence is not negative evidence. SI Appendix, Tables S2 and S10 give the quantitative comparisons and complete branch dispositions.

We analyzed 884 graph-buildable inverted-Alu substrates across four individual GTEx tissues and Combined, a pooled endpoint. Complete pairs were assigned without overlap to 566 training, 139 validation, and 179 test pairs. Discovery combined the published training and validation pairs into pair-disjoint OOF comparisons with pair-clustered uncertainty. For final fitting, the published roles were restored: training fit the selected components, validation fixed checkpoints and thresholds, and the frozen specification was evaluated on the canonical test (Fig. 2). Development evidence therefore attributes branch-level effects; the held-out test evaluates only the final synthesis.

Whole-duplex sequence and structure define a strong but non-universal reference. The search first established what the common reference already contains. The compact sequence–structure model achieved within-context AUROC of 0.869–0.933 across the five human endpoints and 0.823–0.865 across three nonhuman datasets (Fig. 3A). The datasets differ and these ranges do not estimate a common effect. They nonetheless show that whole-duplex sequence and predicted structure predict editing beyond one species or tissue.

Transfer was strongest among human endpoints. It varied more among nonhuman datasets, and human–nonhuman transfer was lower and asymmetric (Fig. 3B and C). Sequence–structure regularities extend across contexts without being universal. These datasets cannot separate the effects of lineage, substrate composition, assay, labeling, and cellular environment.

High discrimination does not imply biological completeness. The reference can rank sites while leaving substrate availability, tissue state, and unmeasured conformations unresolved. Every subsequent branch therefore asks whether a defined information source adds signal beyond this common starting point; predictive accuracy is not treated as explained biological variance.

Genomic organization adds between-substrate information. The substrate-scale branch asked whether genomic organization contributes information absent from the folded graph. Separation between Alu arms correlated negatively with edited-site fraction in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 4B). This distance is absent from the site-level input, which joins the arms with a fixed separator. Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 4C; SI Appendix). Genomic organization therefore contributes reproducible between-substrate signal beyond the tested graph summaries. The modeled outcome is editing propensity; duplex realization and ADAR engagement were not measured.

Editing differs across tissues, but current bulk measurements do not resolve the variation. Editing outcomes varied among tissues even when genomic site identity was fixed. In a post-hoc analysis of development pairs, 1,221 of 74,758 sites observed in at least two individual tissues switched strictly between edited ($\geq 15\%$) and nonedited ($< 1\%$) states. Among 50,162 sites observed in all four tissues, 17.4% spanned at least 15 EditingIndex points. Alu-pair rankings on identical site sets still correlated across every tissue pair ($\rho = 0.727$–0.921 for mean EditingIndex; SI Appendix, Fig. S7). Stable substrate differences coexist with tissue-dependent site outcomes.

Predicted substrate burden did not translate reliably into better site-level classification. A cross-fitted gate added 0.0084 AUPRC but only 0.0041 F1, whose interval included zero, and increased Brier score by 0.0051. A neural burden auxiliary also failed its prespecified criterion. Static geometry captures pair-level propensity without identifying why individual sites depart from it.

The tested contextual measurements added no reproducible residual signal at the site level. They comprised replicated ENCODE eCLIP occupancy, sequence-based RBP predictions, tissue-average abundance and their interactions, with randomized or shuffled controls (19–22). Sequence-based predictions complemented eCLIP, for which Alu multimapping makes peak absence ambiguous.

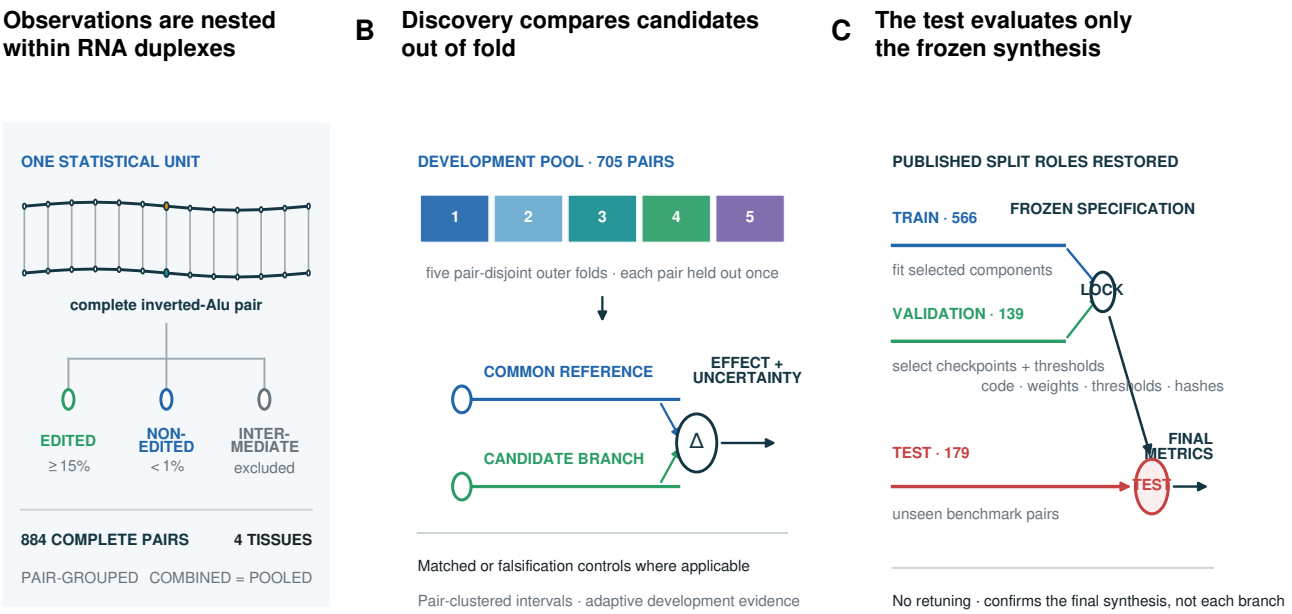


Fig. 2. Benchmark units and evaluation regimes preserve pair independence. (A) Site labels are nested within 884 complete inverted-Alu pairs; intermediate observations are excluded from the canonical task, and Combined is pooled. (B) Discovery pools the published training and validation pairs into five pair-disjoint outer folds. Candidate and reference models use the same folds, relevant matched or falsification controls, and pair-clustered uncertainty. (C) For final fitting, the published split roles are restored and the specification is frozen before evaluation on 179 unseen benchmark pairs. The held-out test evaluates the final synthesis, not each adaptively examined development branch.

Neither cell-line occupancy nor tissue-average abundance reproducibly improved the reference. Tissue identity alone nearly separated cross-tissue label switches, leaving four tissue means unable to identify a transferable RBP-specific effect. This negative result applies to the tested proxies and does not conflict with established RBP regulation (23–26).

Transcript and repeat annotations also failed to improve prediction beyond the graph. Exact-tissue bulk H3K36me3 overlap produced a small liver AUPRC increment without stable AUROC, F1, or pair-level benefit (27, 28). Across these tests, static, cell-line, and tissue-average measurements did not recover the site-level variation left after substrate propensity was modeled. The source of that variation, whether cellular state, measurement noise, finite data, or representation error, remains unresolved.

Pair-first weighting balances dependent observations across substrates. The statistical branch separated a sampling problem from the biological search. The canonical row set contains only part of the available high- and low-editing observations. Under uniform site sampling, a substrate's influence grows with its number of observations even though sites in one duplex share sequence, structure, genomic context, and often editing behavior.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 5A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that expanded pool fixed and drawing a pair before class and site contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). Label recovery and substrate weighting made separable contributions.

The pair-first formulation improved the same-fold reference in all five endpoints (Fig. 5B); all 15 AUROC, AUPRC, and F1 intervals excluded zero. Pair-first sampling is therefore a weighting correction for dependent observations in this benchmark, not evidence for an additional biological determinant.

Pair-first learning improved log loss most for pairs with the fewest eligible sites, whereas the two largest site-count quintiles favored row-wise learning (SI Appendix, Fig. S8B).

Target-centered local context adds information beyond whole-duplex pooling. The target-scale branch asked whether whole-duplex pooling obscures differences among candidate adenosines. Simple target-contact geometry improved ranking and calibration but not F1. Explicitly retaining the focal graph node improved AUROC and AUPRC more consistently than an otherwise matched duplicate-global control. Together, these comparisons localized additional predictive signal near the target.

The selected local-augmented graph predictor uses a 65-position representation centered jointly on the focal adenosine and its dot-bracket-aligned helix partner. A low-capacity dilated convolution encodes nucleotide identity, pairing state, focal and partner indicators, and signed relative position. Its output is fused with the whole-duplex graph embedding before classification. In muscle OOF data, the model reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 5C).

A matched-capacity control retained the same graph-plus-convolution architecture and optimization but shifted

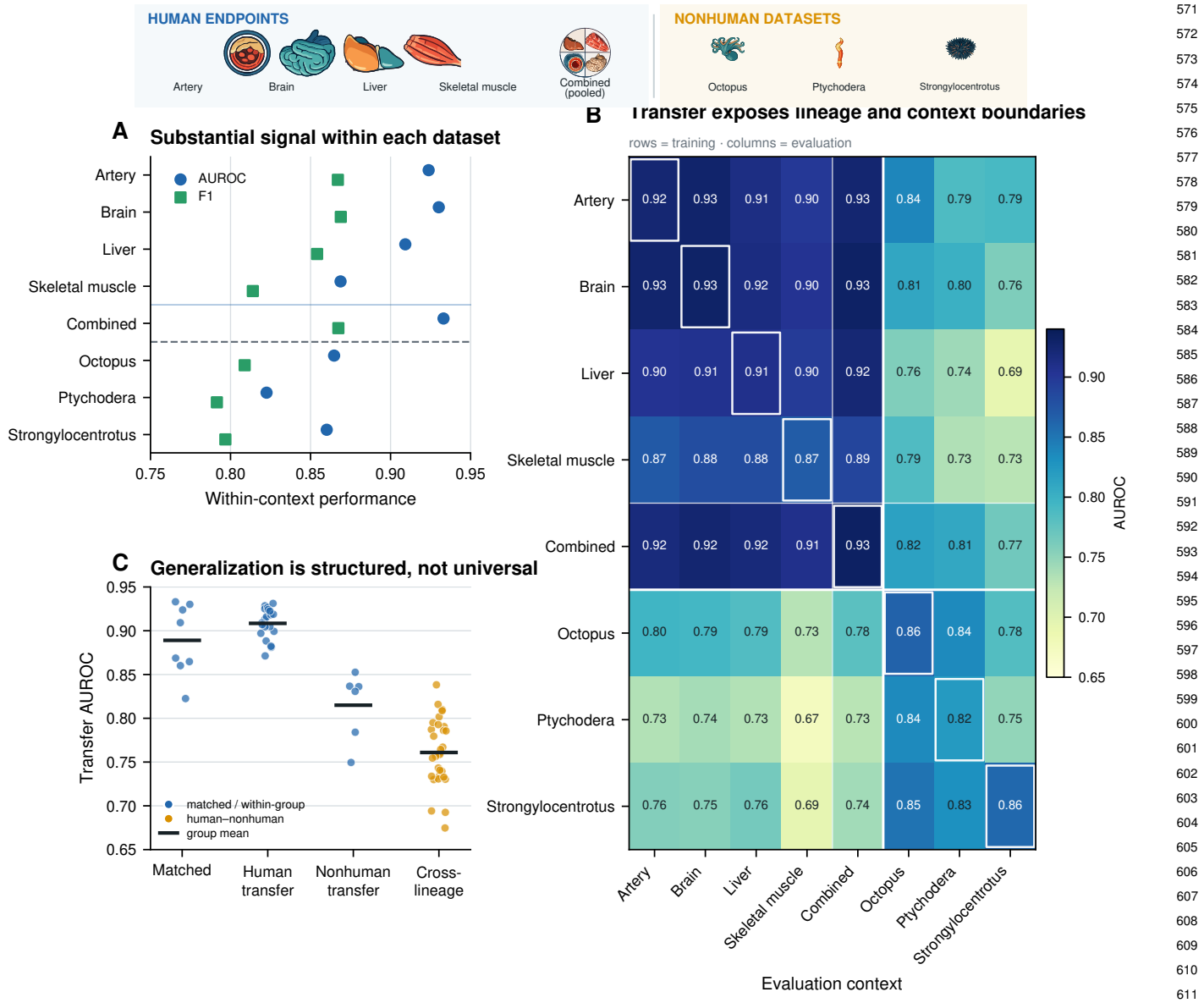


Fig. 3. Sequence and predicted structure carry substantial but non-universal signal. The tissue and lineage illustrations identify dataset classes and encode no measured values. (A) Within-context AUROC and F1 for the retained compact sequence–structure baseline across five human endpoints and three nonhuman datasets. Combined is pooled. (B) AUROC when each training context is evaluated in every context; white lines separate human and nonhuman blocks, and outlined diagonal cells mark matched-context evaluations. (C) Transfer grouped by the relation between training and evaluation contexts; horizontal bars are group means. Dataset heterogeneity precludes causal attribution or meta-analysis.

the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. The gain depends on the correctly centered, pairing-organized neighborhood, not simply on added convolutional capacity. Because the full window was shifted, the control does not isolate the predicted partner from the rest of that neighborhood. Liver replication and the secondary natural-prevalence analysis showed the same pattern.

In a descriptive structural stratification, Brier score improved for paired and unpaired targets and across every target-stem extent (SI Appendix, Fig. S8C and D). The gain cannot be reduced to a binary paired-status annotation or one stem-length threshold.

Whole-duplex and target-local views remain complementary. The whole-duplex and local-augmented predictors made sufficiently different errors to benefit from combination. A fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver. All six pair-clustered intervals excluded zero (Fig. 5D).

The final predictor uses two independently trained pair-first components: a whole-duplex-only graph predictor and a graph predictor augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed 0.5/0.5 mean, with no weight search (Fig. 5E). This fixed combination retains both predictive views without fitting a flexible stacker to only hundreds of independent substrates.

A

Evidence map: which information changes the recoverable signal?

INFORMATION LEVELS	Tested formulation	Outcome	Interpretation
	LEARNING & SUPERVISION		
LEARNING	Pair-first expanded supervision	● Selected	More labels; correct pair weight
	Continuous + ordinal auxiliary labels	▲ Limited	Small ranking gain
	Shared four-tissue training	▲ Limited	Transfer below preset criteria
	TARGET REPRESENTATION		
TARGET-LOCAL	Aligned target + helix-partner view	● Selected	Passes shifted-window control
	Focal node / contact geometry	▲ Limited	Localizes target effect
	Frozen RNA-language embeddings	✖ Not supported	F1-only; no ranking gain
	Substrate-wide all-A decoder	✖ Not supported	Stopped for futility
	Larger / deeper graph networks	✖ Not supported	No consistent benefit
	Graph Transformer / GATv2 expansion	+ Not pursued	Simpler validated model sufficient
	SUBSTRATE & CELLULAR CONTEXT		
SUBSTRATE / CONTEXT	Pair geometry / editing burden	◆ Supported†	Substrate propensity detectable
	Hierarchical engagement gate	▲ Limited	Ranking gain; unstable F1
	Bulk H3K36me3 overlap	▲ Limited	Liver AUPRC only
	RBP binding × bulk expression	✖ Not supported	No residual gain
	Transcript / repeat annotation panel	✖ Not supported	No site-level gain
	Matched donor/cell multimodality	■ Deferred	Matched data unavailable
	Matched in vivo structure / isoforms	■ Deferred	Matched data unavailable
	INTEGRATION		
INTEGRATION	Fixed whole-duplex + local-augmented logits	● Selected	Complementary predictions
	Unified end-to-end architecture	✖ Not supported	Below frozen ensemble

Comparators, effect sizes, and intervals: Table S2.

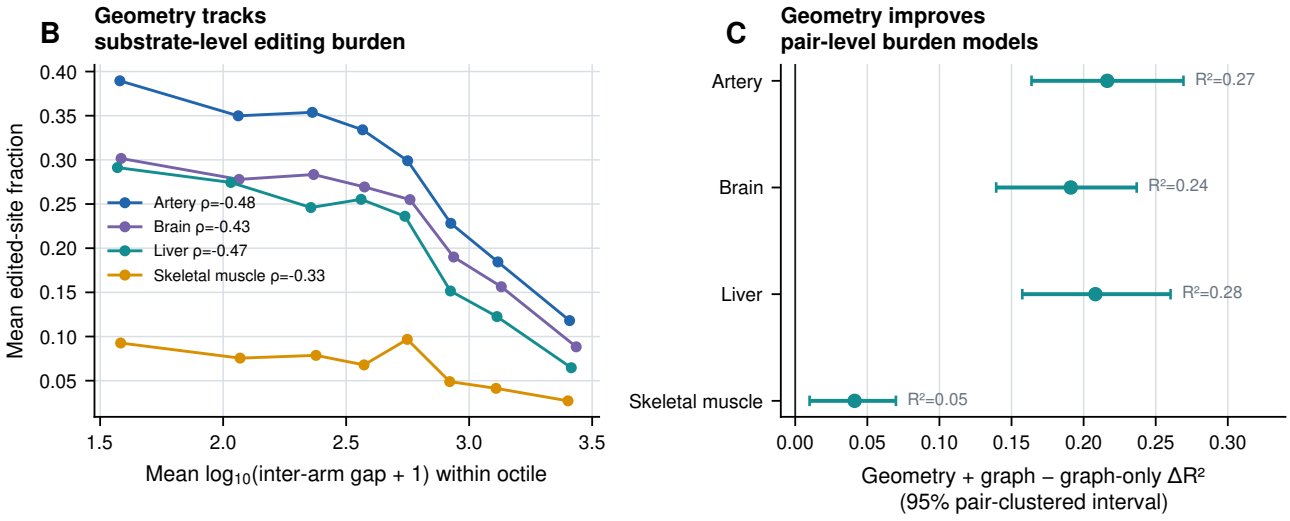


Fig. 4. The evidence map identifies a distinct substrate-level signal. (A) Search outcomes grouped by information scale. SI Appendix, Table S2 gives each comparator, effect size, and interval. "Supported†" denotes a reproducible observation not yet useful for site prediction; "not supported" applies only to the tested formulation, and deferred measurements were unavailable. (B) Mean edited-site fraction across octiles of inter-arm gap in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 after adding genomic geometry to graph summaries. Bars are 95% pair-clustered bootstrap intervals; labels give absolute R^2 for the combined model.

In a post-hoc equal-pair analysis restricted to duplexes containing both labels, the fixed ensemble improved mean within-duplex AUROC over the pair-first component in muscle (+1.78 points; 95% pair-bootstrap interval +1.21 to +2.40) and liver (+1.38 points; +1.06 to +1.70). The aligned-local component alone showed a clear within-duplex increment only in muscle, so this analysis supports complementary within-substrate ranking rather than a replicated local-component effect (SI Appendix, Fig. S8A).

Most tested model-expansion routes yield limited returns. The model-sufficiency search asked whether four common routes to greater modeling power could close the residual gap: richer supervision, cross-context or multitask learning, richer or pretrained representations, and greater capacity or integration. Comparisons within each class used the strongest compatible reference. Expanding the pool of valid high/low labels helped, whereas binary–ordinal–continuous supervision added only a small ranking effect and reduced natural-prevalence performance

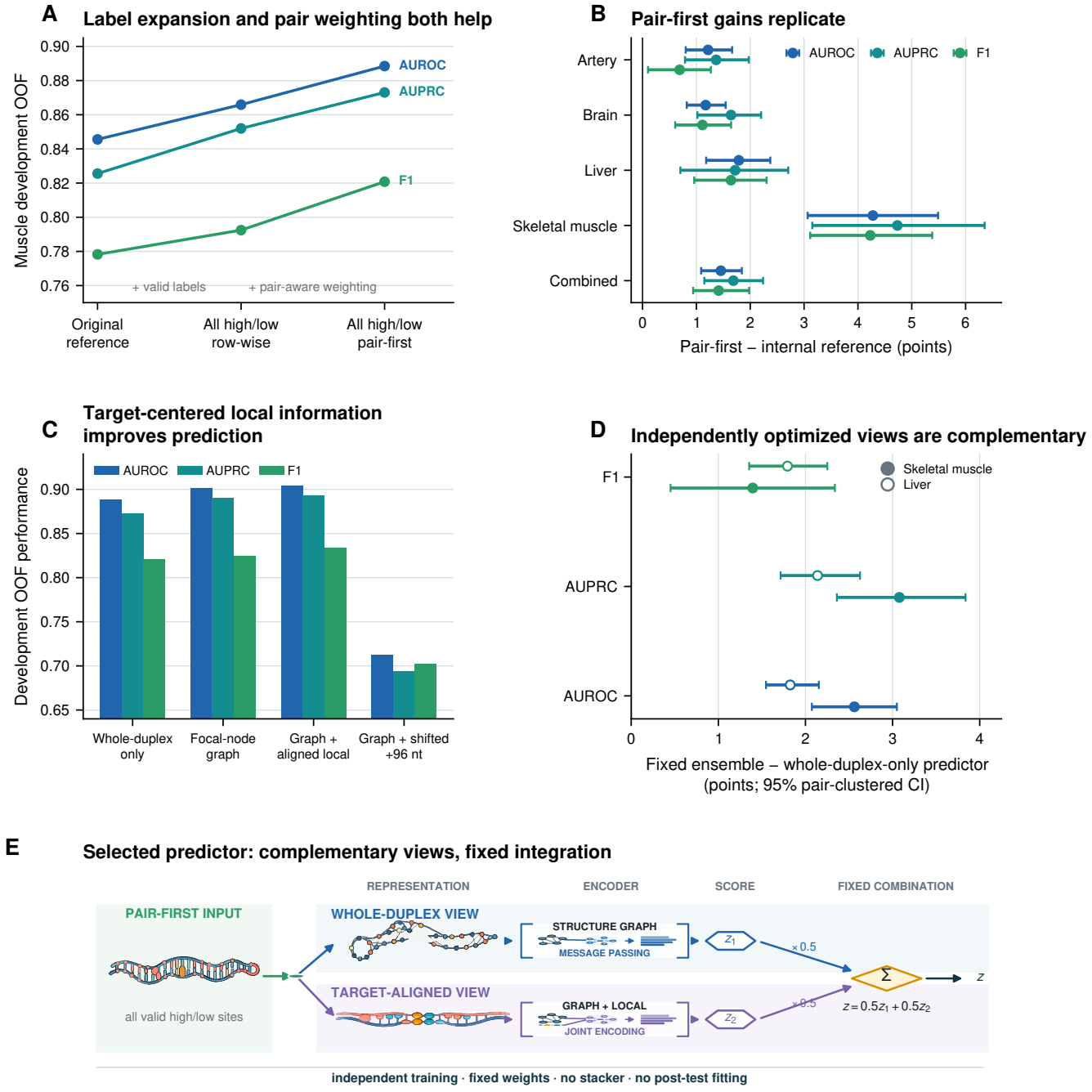


Fig. 5. Pair-first learning and target-centered representation provide complementary gains. (A) Same-fold ablations separate recovery of additional valid high/low labels from pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, focal-state retention, and aligned-local fusion form an evidence ladder; an otherwise identical local model is shifted by 96 nt. (D) A fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) The predictor combines independently trained whole-duplex-only and graph-plus-local models without searched weights or post-test fitting. The RNA illustrations are conceptual. All panels show development evidence.

when combined. Shared four-tissue learning produced limited transfer and worse mean calibration (Fig. 4A; SI Appendix).

Representation changes were similarly selective. Focal-node retention and the aligned local encoder improved the information available at the target, but frozen RiNALMo embeddings did not add ranking signal beyond matched controls (29). Larger graph networks, substrate-wide decoding, engagement-style hierarchical formulations, and the preregistered unified end-to-end model were limited, neutral, or adverse. A fixed prediction-level mean outperformed the tested connected integration. Across these classes, alignment with substrate dependence and target position mattered more than generic model sophistication. This conditional saturation applies to the tested measurements, representations, effective sample size, and evaluation regime; it is not a universal limit on future architectures.

A frozen synthesis improves prediction on unseen benchmark pairs. The final test asked whether principles selected across the search combine on unseen benchmark pairs. All components, checkpoints, thresholds, weights, code, and hashes were frozen before canonical held-out evaluation. The selected predictor improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 6A), with mean gains of 0.0277, 0.0261, and 0.0253 over reproduced supplied-checkpoint AdarEdit. Every endpoint had a positive 95% pair-clustered interval for all three metrics, and Brier score decreased throughout (Fig. 6B and C; SI Appendix, Table S1).

Combined overlaps the individual tissue endpoints, so the mean is descriptive rather than a five-tissue meta-analysis. The frozen combination generalizes to unseen pairs within this benchmark. Its component-level attribution rests on the controlled development comparisons above.

At frozen validation-selected thresholds, the selected predictor corrected 32.0–50.6% of baseline false positives while converting 1.7–5.9% of baseline true positives into false negatives (Fig. 6D). A secondary matched-recall analysis is reported in SI Appendix, Table S9.

Discussion

The resulting empirical map describes endogenous A-to-I editing at the resolution of this benchmark. Whole-duplex sequence and predicted structure provide a strong intrinsic reference. Genomic organization adds between-substrate information, and target-centered context organized by predicted pairing adds signal beyond whole-duplex pooling. Tissue dependence is visible when the same sites are compared across tissues, yet current bulk and cell-line measurements do not identify the state responsible for that variation. Retaining limited and unsupported branches is essential to this interpretation: positive results locate recoverable signal, while controlled null results delimit what the tested measurements do not add.

The substrate result extends earlier observations that distance between oppositely oriented Alus predicts aggregate Alu editing (16). Within a common predictive design, geometry explained pair-level burden beyond graph-derived sequence and structure summaries, but did not become a reliable site-level correction. At the nucleotide scale, prior biochemical and comparative work implicated sequence preference, mismatches, bulges, and helix context (13–15, 17). The aligned-versus-shifted comparison now shows that correctly centered, pairing-organized local context contributes beyond whole-duplex pooling. It does not isolate the predicted partner or establish a molecular mechanism. Pair-first weighting solves a different problem by preventing substrates with more observed sites from dominating estimation.

The context branch exposes the main measurement gap. Perturbation studies establish that RBPs and chromatin-associated processes regulate editing (23–27), and matched genomic sites clearly change across tissues here. Cell-line occupancy, tissue-average abundance, transcript and repeat annotations, and bulk chromatin nevertheless supplied no stable residual site-level signal in the tested formulations. Those null results do not make the underlying biology irrelevant. They show that coarse or unmatched proxies do not resolve regulator exposure, transcript state, or RNA conformation for an individual editing event.

The secondary modeling search reached the same boundary from four directions. Graded supervision and cross-tissue sharing yielded limited gains; generic RNA foundation representations, greater graph capacity, substrate-wide decoding, and connected end-to-end integration did not improve the selected reference. The clearest gains instead matched the structure of the problem: pair-first learning respected substrate dependence, and target-centered encoding preserved local information that global pooling obscured. The fixed ensemble conservatively retains those two views. These experiments do not define a universal ceiling for AI. Under the tested regime, generic model sophistication alone did not close the gap, shifting attention toward information absent from the benchmark. Residual error could reflect donor- or cell-specific ADAR state, transcript processing, regulator occupancy, in vivo conformation, measurement error, finite data, model misspecification, or biological stochasticity. The present data cannot apportion these sources.

The scope remains bounded. The primary benchmark contains graph-buildable human inverted-Alu substrates and tissue summaries, not all editing substrates or donor-resolved measurements. Heterogeneous nonhuman datasets show transfer loss but confound lineage, substrate class, assay, and labeling effects. Predicted secondary structure is a surrogate for the conformations available to ADAR in vivo. The canonical task is balanced and excludes intermediate observations; natural-prevalence analyses are secondary, and eCLIP coverage is cell-line-specific and repeat-censored. Component attribution comes from adaptively selected, nonmultiplicity-adjusted development comparisons. The frozen test assesses unseen pairs from the same benchmark construction, not transportability to an independently generated human cohort (18).

The frozen two-view predictor provides an operational check on the map: principles selected during development improved held-out discrimination, classification, and calibration when combined. Across the major biologically defined scales accessible to this benchmark, the study identifies stable substrate- and target-level signals, a context-dependent remainder that present bulk measurements fail to resolve, and limited returns from generic model expansion. The next decisive data should pair editing with transcript processing, ADAR and interferon state, enzyme localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation in the same donors or cells and substrates.

Materials and Methods

Search design and evidence classification. We organized candidate information into intrinsic whole-duplex or substrate properties, transcript and cellular context, and target-local selection. Dependence among sites from one duplex formed a separate statistical axis. A secondary model-sufficiency axis tested richer supervision, cross-context sharing or multitask learning, richer or pretrained representations, and greater capacity or integration. Branch priority reflected biological rationale, novelty beyond the sequence–structure reference, compatibility with available measurements, leakage risk, and effective sample size. Each valid branch used pair-disjoint development folds, an appropriate same-fold reference, and matched, randomized, location-shifted, or capacity-matched controls where applicable. Selected branches met recorded promotion criteria and

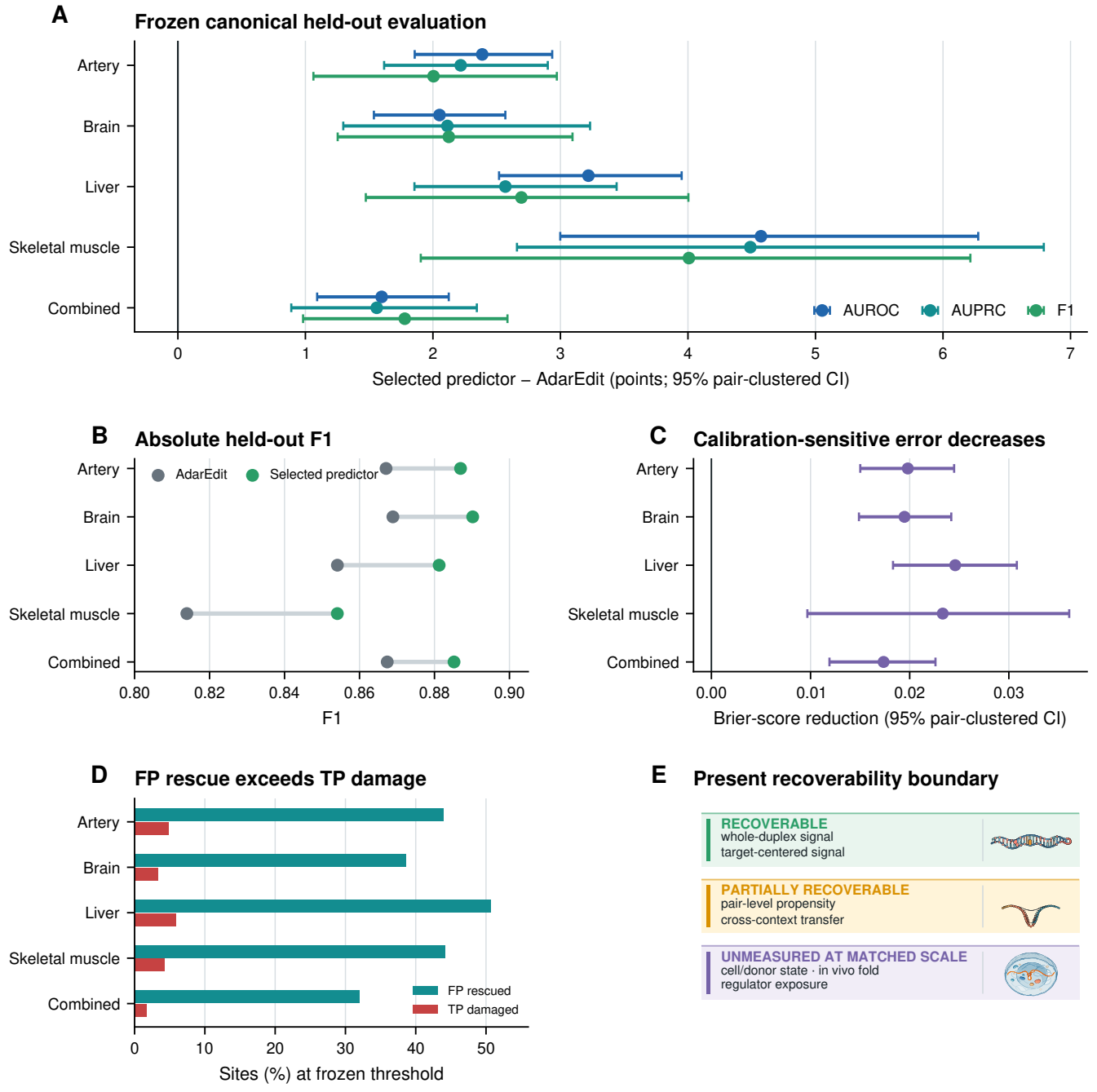


Fig. 6. Frozen evaluation confirms within-benchmark generalization and summarizes what current measurements recover. (A) Selected predictor minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score. (D) Baseline false positives corrected and baseline true positives converted to false negatives at frozen validation-selected thresholds. (E) Information captured, partially resolved, and unavailable at matched biological scale; tier-specific icons are conceptual. Combined is pooled. All quantitative panels use the frozen canonical held-out evaluation.

entered the final synthesis; supported findings were reproducible but not selected; limited findings were metric- or context-specific. Not supported denotes failure of the tested formulation, deferred denotes unavailable matched data, and superseded denotes a direction not pursued after stronger lower-complexity evidence emerged. Development comparisons were adaptive and were not treated as multiplicity-adjusted confirmatory tests. Only the frozen final synthesis entered the canonical held-out evaluation. SI Appendix, Tables S2 and S10 report the branch-level evidence and disposition.

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human sites were labeled edited at EditingIndex $\geq 15\%$, nonedited at EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. We verified pair identities and the fixed 566/139/179 train/validation/test assignment before analysis. Combined was used only as its published pooled endpoint.

Development protocol. All discovery used the original train and validation pairs, divided into 5 outer folds by pair ID. Learned preprocessing, feature selection, checkpoint choice, and threshold choice excluded the outer fold. We concatenated predictions only after every row had been scored by a model that was not trained on its pair. The canonical test remained inaccessible during development. Development OOF metrics estimate same-fold incremental effects; supplied-checkpoint test metrics remain the authoritative reference.

Models. The first component retained the compact AdarEdit graph attention encoder and was trained from the expanded high/low site pool using uniform pair-first and equal-class sampling. The second component was trained independently with the same graph encoder. It added a 65-position target/helix-partner-aligned representation encoded by two dilated convolutional layers, then fused the local and graph embeddings before prediction. The selected probability was $\sigma[(\operatorname{logit} p_{\mathrm{whole}} + \operatorname{logit} p_{\mathrm{augmented}})/2]$, with weights fixed at 0.5/0.5 and no weight search. The shifted-window control preserved the graph-plus-convolution architecture and optimization while moving the local center 96 nt. SI Appendix gives full feature definitions, architectures, ablations, and biological-context protocols.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each frozen training run. For uncertainty, a pair-clustered bootstrap with 2,000 replicates resampled Alu pairs and retained all their sites. Development intervals quantify paired effect uncertainty and were not treated as multiplicity-adjusted confirmatory tests. Selection also required recorded decision criteria and controls. Code, inputs, weights, epochs, thresholds, and checkpoint SHA-256 values were frozen before one held-out evaluation; no subsequent model selection or tuning occurred. Natural-prevalence and matched-recall analyses are secondary and described in SI Appendix.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. GPT Image 1.5 generated only label-free, nonquantitative figure components; prompts, candidate selection, and hashes are disclosed in the repository. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

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